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Green solvents as a sustainable strategy to reduce the impact of hazardous waste on human health

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Abstract
Green solvents have become indispensable in advancing sustainable chemistry by addressing the environmental, health, and safety concerns associated with traditional volatile organic compounds. These solvents, often derived from renewable resources or designed for reduced toxicity and improved biodegradability, align with the principles of green chemistry and offer promising alternatives in both academic research and industrial applications. The transition to greener solvent systems is not only driven by regulatory pressure but also by the increasing demand for cleaner technologies in pharmaceutical, agrochemical, polymer, and food sectors. This review provides a comprehensive overview of the major classes of green solvents including bioalcohols (ethanol, butanol), terpenes (limonene, pine oil), fatty acid methyl esters (soy and palm-based FAMEs), natural deep eutectic solvents, and next-generation solvents such as γ-valerolactone and Cyrene™. The physicochemical properties, functional advantages, and environmental profiles of these solvents are critically examined using established greenness evaluation tools such as the CHEM21 selection guide and solvent selection frameworks. The review further explores innovative case studies and technologies demonstrating the application of green solvents in extraction, synthesis, biopolymer recycling, pesticide formulation, and detergent development. Comparative data, performance benchmarks, and sustainability metrics are provided to illustrate their efficacy relative to conventional solvents like DCM, or hexane. Overall, the review highlights how green solvents contribute to public health, circular economy goals, reduce hazardous waste, and enhance process safety, thereby reinforcing their role as transformative agents in the global shift toward more sustainable and responsible chemical manufacturing practices.

Keywords Green solvents, Sustainable chemistry, Bio-based solvents, NADES, Solvent selection guides, Green extraction, Solvent recycling, Environmental impact

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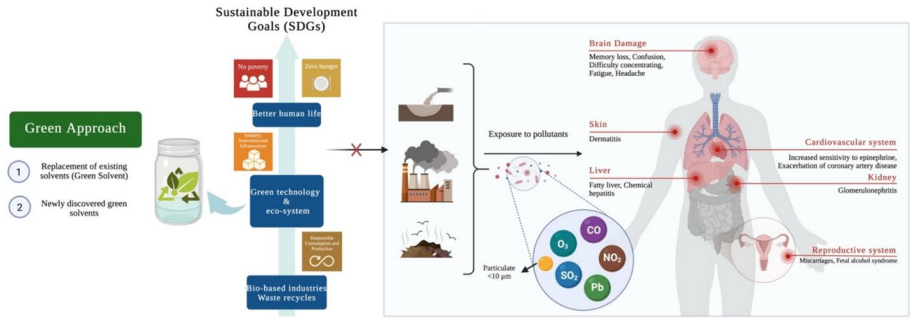
Graphical Abstract



1 Introduction

Solvents are indispensable in the chemical, pharmaceutical, agricultural, and manufacturing industries, serving as reaction media, extraction agents, formulation components, and cleaning agents. Globally, millions of tons of solvents are consumed annually, contributing significantly to volatile organic compound (VOC) emissions, occupational hazards, and environmental pollution. Traditional organic solvents often derived from fossil resources pose critical risks to human health and the environment due to their toxicity, flammability, and persistence in ecosystems. In response to these concerns, the principles of green chemistry have gained global momentum, with international frameworks such as the United Nations Sustainable Development Goals (UN SDGs) emphasizing the need for sustainable industrial practices. Within this context, green solvents have emerged as a critical area of innovation, offering the potential to reduce environmental burden without compromising industrial performance.

The transition toward green solvents is not solely a matter of environmental compliance but also of economic and public health significance. Exposure to conventional solvents has been associated with respiratory disorders, carcinogenicity, endocrine disruption, and groundwater contamination. As solvent-intensive sectors such as pharmaceuticals and agrochemicals expand, the need for safer alternatives becomes increasingly urgent. Regulatory agencies, including the U.S. Environmental Protection Agency (EPA) and the European Chemicals Agency (ECHA), have responded by tightening restrictions on hazardous solvents. These evolving policies have encouraged industries to adopt environmentally benign, biodegradable, and renewable solvent systems that align with health and safety goals while enabling circular economy models.

Over the past two decades, research on green solvents has evolved from conceptual development to real-world application. Novel solvent classes such as Natural Deep Eutectic Solvents (NADES), bioalcohols (e.g., ethanol, butanol), ionic liquids, terpenes (e.g., limonene, pine oil), fatty acid methyl esters (FAMEs), and bio-based lactones (e.g., γ-valerolactone, GVL) have demonstrated promising physicochemical and functional properties. These green solvents have shown compatibility with synthetic chemistry, pharmaceutical formulations, natural product extraction, biocatalysis, and polymer recycling. In particular, NADES systems customizable mixtures of natural components have emerged as efficient extractants for phytochemicals with minimal toxicity and high biodegradability [81, 86]. The commercial interest in such technologies is evidenced by the increasing number of green solvent-related patents and industry-academic

partnerships globally [160]. Despite these advances, there remains a lack of integrative reviews that evaluate green solvents across multiple application domains while incorporating industrial developments, regulatory classifications (e.g., CHEM21), and sustainability metrics. Most existing studies are fragmented, focusing either on chemical synthesis or extraction, with limited emphasis on holistic solvent performance and public health relevance. This review addresses that gap by offering a comprehensive analysis of green solvent types, properties, applications, and innovations across multiple sectors. It also examines public health implications, industry case studies, and future perspectives. By synthesizing evidence from recent literature, industrial patents, and regulatory guidelines, this paper aims to support researchers, formulators, and policymakers in making informed decisions that foster sustainable solvent use.

1.1 Principles of green chemistry

Green chemistry, conceptualized by John Warner and Paul Anastas in 1998, is grounded in a framework of 12 guiding principles (Fig. 1) [12]. This scientific philosophy emphasizes the intentional design of chemical products and processes that actively reduce or eliminate the generation and use of hazardous substances. Its fundamental objective is to mitigate waste production and minimize the reliance on toxic solvents, particularly within industrial and analytical applications, thereby safeguarding both human health and the environment [10, 13, 112]. By adopting green chemistry practices, risks to workers and ecosystems are substantially minimized throughout the entire lifecycle of a product, from synthesis to disposal [115, 156].

The initial focus of green chemistry was to transition chemical practices from unsustainable methods toward more sustainable and environmentally responsible alternatives

Fig. 1 12 Principles of green chemistry.
Source: Adopted from Armenta et al. [16]

[11, 130]. A variety of initiatives have since emerged under the umbrella of "green chemistry" [45, 47], including prominent programs led by the US Environmental Protection Agency (EPA). These initiatives are designed to foster innovative technologies that substantially reduce hazardous waste and lessen environmental harm. A notable example is the EPA's Green Chemistry Challenge Awards, which celebrate advancements that embed green chemistry principles into the design, production, and application of chemicals [12]. Building on this framework, Anastas et al. extended the 12 principles of green chemistry to the field of analytical science, giving rise to the concept of "green analytical chemistry" This approach aims to minimize the generation and use of hazardous substances at every stage of the analytical process, prioritizing safer chemical design, renewable raw materials, and increased energy efficiency [10, 54, 146, 156, 169]. In 2001, Namieśnik et al. further refined this concept by identifying four key criteria for green analytical chemistry: reducing the consumption of organic solvents and reagents, lowering waste generation, eliminating highly toxic and ecotoxic substances, and decreasing energy use in analytical procedures. Their contributions have spurred developments such as solvent-free extraction techniques and the adoption of safer reagents, both of which are now widely applied in laboratories to reduce their environmental impact [12, 185].

An essential component of green chemistry is the incorporation of environmentally benign solvents, commonly referred to as "green solvents" (Fig. 2). One of the earliest systematic approaches was introduced by Curzons et al. [51], who proposed solvent selection guides based on environmental, health, and safety (EHS) criteria, emphasizing the need for holistic assessments beyond performance. Their work laid the foundation for evaluating solvents based on sustainability metrics. Capello et al. [39] later contributed significantly by formalizing the quantitative assessment of solvent greenness through a scoring methodology that integrated life cycle impact, hazard potential, and operational factors. Thus, while Capello et al. advanced the quantification of solvent greenness, the conceptual basis for green solvent selection was already established in the late 1990s. Since then, various strategies related to green solvents have been developed. These include the adoption of biosolvents derived from oleochemicals [170], the

Fig. 2 Types of green solvents

replacement of hazardous organic solvents with safer, biodegradable, and environmentally friendly alternatives that pose lower risks to human health, the ozone layer [51, 52, 101, 128],and the utilization of supercritical fluids [22, 140, 147] and low vapor pressure ionic liquids [116, 171], both of which offer greener solvent options.

Duarte et al. further expanded on green chemistry strategies by suggesting a range of practices, such as modifying sample preparation techniques, optimizing reaction conditions, employing alternative solvents and feedstocks, and even directly analysing untreated samples to reduce solvent use and energy consumption [58]. Nonetheless, for a solvent to fully align with the principles of green chemistry, it must demonstrate high solvency power, low environmental impact, good selectivity, low toxicity, recyclability, and biodegradability and should be economically derived from renewable resources. Despite these criteria, finding a solvent that meets all these requirements remains a significant challenge. Consequently, solvent selection often involves balancing multiple factors, including the extraction technique, analytical methodology, sample matrix, and characteristics of the target analytes [48].

In defining what constitutes a green solvent, Jerome and Gu proposed a comprehensive framework that goes beyond basic toxicity or biodegradability. According to their analysis, an ideal green solvent must exhibit several critical properties: low toxicity to human health and the environment, high biodegradability, renewable sourcing, low volatility, favorable energy efficiency during production and recovery, and minimal impact on climate change and ozone depletion. These criteria also emphasize solvent recyclability, compatibility with green reaction conditions, and economic viability at scale. The integration of these multidimensional considerations ensures that the selection of solvents is not merely based on replacing one compound with another, but rather guided by a holistic evaluation of lifecycle sustainability. Figure 2 presents a classification of widely accepted green solvents based on these guiding principles [82].

1.2 Environmental pollution caused by organic solvents

The extensive use of solvents in industrial operations contributes significantly to environmental pollution due to the large volumes of waste generated throughout various production processes. Key sectors within the chemical industry, namely, pharmaceuticals, specialty chemicals, and bulk or commodity chemicals, differ in terms of the scale and nature of waste produced during manufacturing [199]. Solvents play crucial roles across a wide range of industries, including dry cleaning, food processing; plastics and rubber manufacturing; metal degreasing and cleaning; production of aviation fuel additives; automotive applications; and the formulation of lacquers, varnishes, paints, adhesives, textiles, printing inks, and pharmaceuticals.

A variety of solvent classes are commonly used in these sectors, such as alcohols (e.g., amyl, butyl, ethyl, and isopropyl), esters (e.g., amyl acetate, butyl acetate, and ethyl acetate), ketones (e.g., methyl ethyl ketone and acetone), petroleum-derived hydrocarbons (e.g., toluene, xylene, and benzene), chlorinated hydrocarbons (e.g., ethyl chloride and methylene chloride), and glycol ethers [143]. The widespread application of many conventional organic solvents, particularly those with high vapor pressure and low biodegradability, contributes to significant emissions of volatile organic compounds (VOCs) and hazardous air pollutants (HAPs), exacerbating environmental and human health concerns.

Industrial waste streams are rich in a variety of contaminants, including heavy metals, toxic chemicals, and organic pollutants [148]. For example, the steel industry generates sludge, whereas the chemical, paper, and mining sectors release harmful chemical residues. Food processing plants contribute biochemical oxygen demand (BOD) pollutants, conventional and nuclear power plants discharge thermal pollutants, and mining operations often cause pH fluctuations in nearby ecosystems [143]. Among these, pharmaceutical manufacturing is recognized as a major contributor to waste pollution, particularly because of its reliance on multistep liquid-phase batch processes [199]. During the synthesis of active pharmaceutical ingredients (APIs), various intermediates are generated, isolated, and purified via the use of large volumes of organic solvents. In a study by Jiménez-González et al. [94] reported that solvent usage can account for up to 80% of the total waste produced in the life cycle of API production, excluding water.

The link between hazardous organic solvents and the principles of green chemistry is well documented in the work of [97]. The increasing awareness of the toxicological and environmental impacts of volatile, persistent, and non-biodegradable solvents such as methanol, acetone, ethyl acetate, and dichloromethane has spurred both industry and academia to adopt greener alternatives. Organic solvents are typically composed of carbon and hydrogen atoms, and may include heteroatoms such as oxygen, nitrogen, or halogens that influence reactivity, polarity, or environmental behavior, but not necessarily volatility. Instead, volatility depends largely on molecular weight, vapor pressure, and intermolecular forces. The volatility of these compounds facilitates their escape into the atmosphere, where they participate in chemical transformations and contribute to secondary pollution.

Among the most significant environmental effects of conventional solvent emissions is their role as precursors to ground-level ozone (O_3). In the presence of nitrogen oxides and sunlight, VOCs undergo photochemical reactions that lead to ozone formation, which has been linked to respiratory diseases, cardiovascular stress, crop damage, and forest decline [134]. Ozone is a potent oxidizing agent, and its formation in the troposphere is a major indicator of poor air quality [20, 99, 124, 132, 154]. In this regard, solvent emissions do not directly correlate with lead (Pb) or thermal pollution, but they contribute significantly to ozone and smog formation, which are critical environmental and regulatory concerns [69, 90].

In recent years, attention has also been given to per- and polyfluoroalkyl substances (PFASs), a class of synthetic organic compounds widely used in industrial applications and consumer goods [80]. Alarmingly, PFAS contamination has been detected in public and private drinking water sources across the United States, raising serious national and global health concerns [179]. Although PFAS are not conventionally used as solvents in manufacturing or chemical synthesis, a few fluorinated PFAS compounds have been employed in niche applications, such as high-performance electronics and surfactant formulations [110]. The mention of PFAS here is not to equate them with common industrial solvents, but to illustrate the shared concerns around environmental persistence, bioaccumulation, and toxicity, which highlight the broader challenge of managing organic pollutants with long-term ecological footprints [65, 179, 183, 186]. As such, while solvents and PFAS differ in structure, function, and prevalence, they both underscore the importance of adopting sustainable chemical practices in industrial systems.

1.3 Effects of organic solvent exposure on humans

Synthetic organic solvents can pose significant health risks to humans, primarily through inhalation, dermal contact, or accidental ingestion in occupational and environmental settings. Although concerns over bioaccumulation are more relevant for certain persistent and hydrophobic compounds (e.g., pesticides or specific siloxanes), most common solvents do not bioaccumulate significantly in biological tissues. Instead, the primary risk associated with organic solvents arises from their volatility, lipophilicity, reactivity, and ability to interfere with central nervous system function and organ systems following acute or chronic exposure. Organic solvents such as carbon tetrachloride (CTC), trichloroethane (TCA), and trichloroethylene (TCE) have been implicated in groundwater contamination, especially in proximity to industrial discharge zones [83]. While these solvents are not pesticides, their structural similarity to other halogenated organics and environmental mobility has prompted regulatory classification. For instance, the U.S. Environmental Protection Agency (USEPA) categorizes CTC as a Group B2 probable human carcinogen (USEPA 2010).

Toxicologically, the absorption, distribution, and impact of many solvents in the human body are influenced by their lipophilicity, which facilitates diffusion across lipid membranes and accumulation in lipid-rich tissues like the brain. However, this does not apply universally. Methanol, for example, is a small, hydrophilic alcohol that does not accumulate in fat-rich tissues. Instead, its acute toxicity arises from its hepatic metabolism into formaldehyde and formic acid, which can cause optic nerve damage, metabolic acidosis, and even death in high doses [119]. This highlights that both lipophilic and hydrophilic solvents can exhibit serious toxicity, albeit through different mechanisms. Lipophilic solvents, including toluene, trichloroethylene, and benzyl chloride, are more commonly associated with central nervous system effects, ranging from dizziness and euphoria to chronic neurotoxicity. These compounds often undergo biotransformation into more hydrophilic metabolites for renal excretion. However, some metabolites may be more toxic than the parent compound. For example, chloroform biotransformation yields phosgene, a potent hepatotoxin and nephrotoxin [59], anxiolytic effects (e.g., toluene) [27], convulsant activity (e.g., fluorothyl) [68], anticonvulsant properties (e.g., toluene) [49], and narcotic and mood-altering effects (e.g., benzyl chloride).

Extensive research has been dedicated to investigating the health impacts of organic solvent exposure, given its well-documented association with a range of acute and chronic health conditions (Figs. 3, 4) [56]. Acute exposure predominantly affects the central nervous system (CNS), with clinical manifestations ranging from headaches, dizziness, and nausea to more severe outcomes such as seizures, unconsciousness, and, in extreme cases, death [197]. Exposure to solvent mixtures frequently results in mucosal irritation, particularly of the eyes, nose, and throat [43]. Alarmingly, organic solvent misuse persists as a public health concern, especially among marginalized youth who seek intoxication through substances such as glues, lighter fuels, and other household chemicals [64]. Volatile substance abuse, particularly involving butane gas and aerosols, has been implicated in approximately 50–60 deaths per year in the UK, with a significant proportion involving adolescents (Advisory Council on the Misuse of Drugs [4, 61]. In occupational settings, solvent misuse has also been reported among certain workers exposed to solvents regularly [129]. Additionally, poor industrial practices and inadequate recognition of solvent volatility continue to contribute to dangerous exposures in

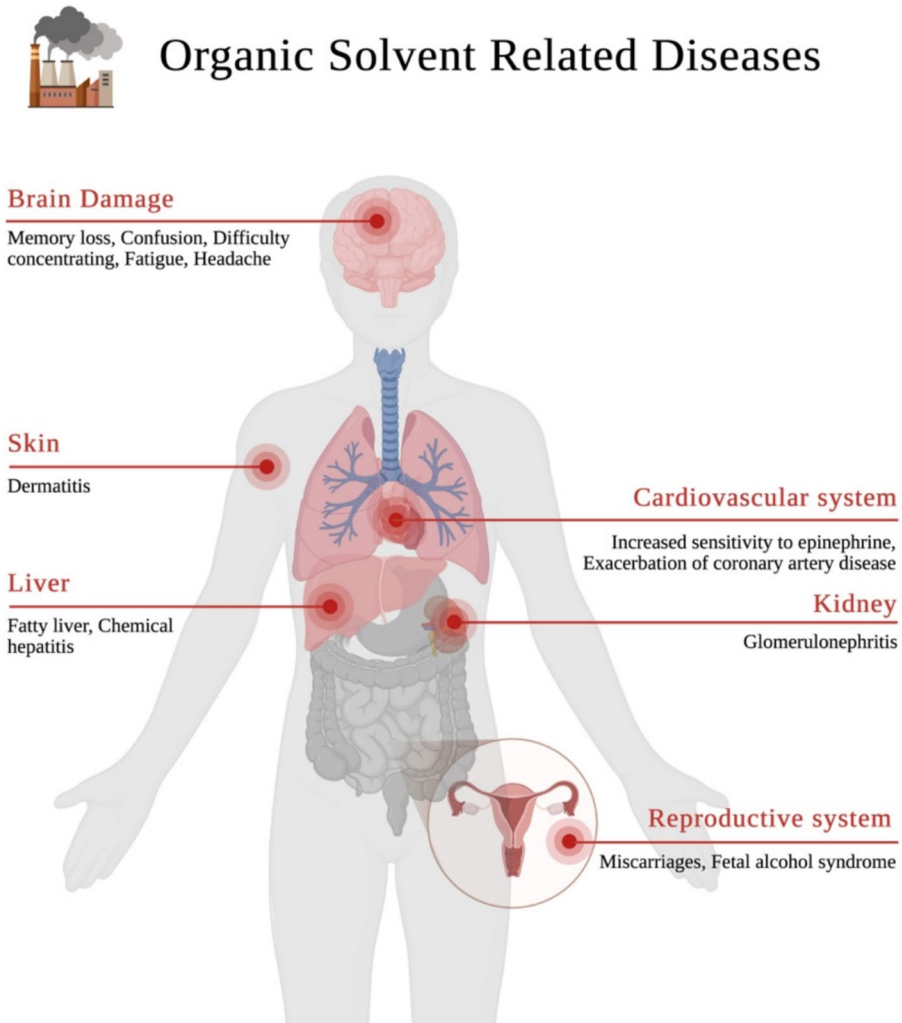


Fig. 3 Health effects of organic solvent exposure on the human body

enclosed spaces, increasing the risk of fires and explosions. Numerous incidents involving the accumulation and ignition of solvent vapours, particularly in confined spaces or poorly ventilated areas, have resulted in fires, severe injuries, and fatalities in industrial settings [144, 150, 195]. These realities underscore the urgent need for increased awareness regarding both the acute toxicological risks and the occupational hazards associated with organic solvents, reinforcing the necessity for strict preventive and safety protocols in workplaces and vulnerable communities.

Long-term exposure to many organic solvents has been linked to serious diseases, including leukemia [182], scleroderma, and renal cancer [31]. The causal relationship between benzene exposure and leukemia is well established,however, the chronic effects of solvents on the nervous system remain an area of ongoing debate [73]. Early investigations introduced the concept of chronic toxic encephalopathy (CTE), which is characterized by symptoms such as memory impairment, personality alterations, and broader neurological deficits. A notable study by Berr et al. [26] further explored this connection, suggesting that prolonged, high-level solvent exposure may contribute to subtle cognitive decline later in life, particularly affecting attention span, memory, and visuospatial function.

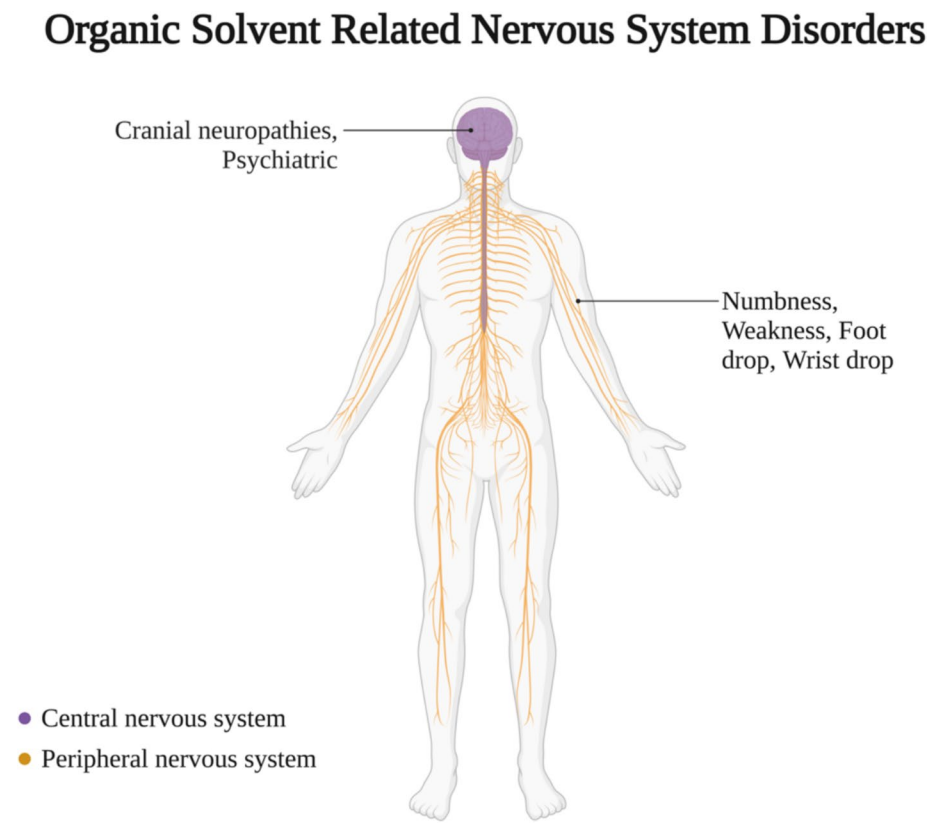


Fig. 4 Health effects of organic solvent exposure on the nervous system

The study conducted within the GAZEL cohort of the French national gas and electric utility workforce, analysed the occupational solvent exposures and cognitive performance of nearly 5000 employees aged between 55 and 65. Cognitive assessments were performed via the Mini-Mental State Examination and the Digit Symbol Substitution Test (DSST). The results revealed a significant association between prolonged exposure to solvents such as benzene, chlorinated solvents, aromatic solvents, and petroleum-derived solvents and an increased risk of cognitive impairment. Importantly, this relationship persisted even after adjusting for variables such as educational attainment, occupational grade, and other health-related factors. These findings strongly suggest that chronic occupational solvent exposure may contribute to cognitive decline during late midlife.

Several studies employing systematic reviews and meta-analyses have significantly enhanced our understanding of the health effects associated with organic solvent exposure, providing critical evidence on this subject. For example, Xiao et al. conducted a systematic review and meta-analysis of 24 relevant studies to examine the link between occupational exposure to organic solvents and breast cancer risk. These findings revealed a statistically significant association, indicating that workers exposed to organic solvents had a 1.18-fold increased risk of developing breast cancer. Subgroup analyses further demonstrated heightened risks, with postmenopausal women and occupationally exposed workers showing 1.35-fold and 1.21-fold increased risks, respectively, thereby reinforcing the correlation between solvent exposure and the incidence of breast cancer [200].

In another comprehensive investigation, Vlaanderen et al. [194] explored the relationship between occupational benzene exposure and the risk of developing chronic myeloid leukemia (CML). This meta-analysis incorporated data from multiple cohort studies, each carefully assessed for quality, considering confounding factors and follow-up durations. The pooled results confirmed a positive association between workplace benzene exposure and an elevated risk of CML. Additionally, Attarchi et al. [17] conducted a study demonstrating that prolonged occupational exposure to mixed organic solvents may contribute to the development of acquired color vision impairment. These findings further highlight the diverse and potentially severe health risks associated with long-term solvent exposure in industrial settings.

Ultimately, the adverse health impacts of organic solvent exposure are well documented. Although the implementation of occupational safety regulations and risk management strategies has been pivotal in reducing exposure levels, residual risks remain. Therefore, to achieve more sustainable and health-conscious industrial practices, transitioning toward the use of "green solvents" presents a viable and increasingly recommended alternative to conventional hazardous solvents.

1.4 Overview of green solvent classes

To rationally replace hazardous solvents in industrial and laboratory practices, it is essential to understand the different classes of green solvents available. Green solvents are broadly defined by their reduced toxicity, lower environmental impact, and improved sustainability metrics as outlined by frameworks such as the CHEM21 guide and the twelve principles of green chemistry. The following are representative categories and key examples:

Bio-based ethers (e.g., 2-Methyltetrahydrofuran, 2-MeTHF), derived from renewable biomass, 2-MeTHF is a promising alternative to tetrahydrofuran (THF) and dichloromethane (DCM). It exhibits lower toxicity, improved safety profile, and enhanced extraction capabilities, particularly in natural product and lipid extractions [28, 77].

Lactones (e.g., γ-Valerolactone, GVL), GVL is produced from levulinic acid and offers excellent biodegradability and low volatility. It has been successfully applied in plant extractions and catalysis, often outperforming traditional solvents like DCM and ethanol in terms of yield and selectivity [42].

Deep Eutectic Solvents (DES) and Natural Deep Eutectic Solvents (NADES), these solvents are formed by mixing hydrogen bond donors and acceptors, typically choline chloride and natural metabolites (e.g., sugars, amino acids, organic acids). They are non-volatile, biodegradable, and tunable for selective extractions. NADES have demonstrated high extraction efficiencies, particularly for polyphenols and flavonoids [53].

Dihydrolevoglucosenone (Cyrene™), a bio-derived dipolar aprotic solvent, Cyrene™ is considered a green substitute for DMF and NMP. It exhibits favorable toxicity and biodegradability profiles and has shown strong performance in synthetic reactions such as peptide couplings [37].

Dimethyl Isosorbide (DMI), DMI is a biocompatible and thermally stable solvent derived from glucose. It is used in pharmaceutical formulations, cosmetics, and polymer applications due to its high solvating power and excellent safety profile [166, 190].

1.5 Industry-specific green metrics

In industrial contexts, a solvent is considered "green" on the basis of its environmental performance. Several methodologies have been developed to assess a solvent's environmental friendliness, including life cycle assessment (LCA), carbon dioxide emission evaluations, and solvent selection tools such as the CHEM21 selection guide [158]. Among these, LCA has garnered significant attention because of its comprehensive approach, which considers potential environmental impacts and resource consumption throughout a product or service's entire life cycle from raw material extraction to end-of-life disposal [127].

[174] was among the first to introduce key green chemistry metrics in the 1990s, notably the E-factor and atom economy, both of which have since gained widespread acceptance. These metrics have been further refined to account for system boundaries [164], and additional mass-based indicators, such as process mass intensity (PMI) [95, 96] and reaction mass efficiency (RME) [164], have been proposed to enhance the evaluation of process greenness. However, while mass-based metrics provide useful insights into material efficiency, a truly comprehensive sustainability assessment requires the integration of environmental impact assessments such as LCA with economic feasibility considerations [50, 70, 102, 108, 189]. In addition, Jessop [91] strongly advocated for the use of LCA in evaluating the environmental impacts of solvents, emphasizing that solvent greenness can vary depending on specific application contexts. For example, factors such as distillation energy requirements or purification processes can influence a solvent's overall environmental footprint. As a result, conducting application-specific LCAs is essential for generating actionable insights and promoting informed decision-making in the pursuit of sustainable chemical manufacturing.

1.6 Green chemistry strategy to achieve sustainable development goals

In 2015, the United Nations introduced the 17 Sustainable Development Goals (SDGs) as a global framework aimed at eradicating poverty, safeguarding the environment, and fostering peace and prosperity by 2030. These goals serve as a blueprint to promote sustainable social, economic, and environmental development worldwide. The distinction between green chemistry and sustainable chemistry lies primarily in their scope. Green chemistry, as defined by Anastas and Warner, focuses on the design, production, and application of chemicals that inherently minimize the generation of hazardous substances [12]. In contrast, sustainable chemistry incorporates the principles of green chemistry while expanding the perspective to include process efficiency, material utilization, energy consumption, and economic viability within its framework [127].

According to the American Chemical Society (ACS) [2], green chemistry is directly aligned with seven core SDGs: affordable and clean energy (SDG 7),good health and well-being (SDG 3); zero hunger (SDG 2); clean water and sanitation (SDG 6); responsible consumption and production (SDG 12); industry, innovation, and infrastructure (SDG 9); and climate action (SDG 13). Through green solvent research, chemical hazards can be mitigated by designing chemicals and processes with reduced toxicity and environmental impact, thus addressing both human health and sustainability challenges. Innovations in green solvents contribute to the development of safer products, reduce resource consumption, and support global efforts toward responsible production and sustainable industrial practices.

Emerging studies further substantiate the critical role of green chemistry in advancing the SDGs. For instance, Kobayashi and Nakajima [103] emphasized the importance of utilizing advanced materials sourced from industrial waste and biomass to foster sustainable development [25, 88, 131]. The recycling and valorization of industrial waste help reduce resource depletion and minimize environmental pollution, directly supporting SDG 12 (Responsible Consumption and Production). Additionally, the exploitation of biomass-derived materials, such as cellulose and related chemicals, contributes to climate action goals (SDG 13) and renewable energy initiatives (SDG 7), thereby promoting a low-carbon and resource-efficient economy [92, 103, 138, 139, 178]. The integration of these sustainable materials into various industries also strengthens innovation and infrastructure (SDG 9), facilitating the growth of a resilient and sustainable industrial sector. Collectively, these practices bridge environmental stewardship with economic advancement, contributing to a more sustainable and prosperous society.

In a study by Fenibo et al. [67] highlighted sustainable agriculture as a sector where green chemistry can significantly impact the achievement of multiple SDGs. Sustainable agriculture is closely connected to green chemistry principles, particularly through the promotion of biodegradable agrochemicals, reduced environmental impact, and enhanced resource efficiency [126]. Biopesticides, for example, represent eco-friendly alternatives to conventional pesticides and are pivotal in achieving SDG 2 (Zero Hunger) by improving agricultural productivity while protecting ecosystems. They further contribute to SDG 1 (no poverty) by enhancing food security and farmer incomes and to SDGs 6, 12, 14, and 15 by fostering responsible resource use and biodiversity conservation [3, 193]. Moreover, green chemistry innovations promote the adoption of cleaner technologies and renewable energy sources, aligning with SDGs 7 and 9. Therefore, embedding green chemistry practices within agriculture and related industries ensures a balanced approach to environmental protection, economic development, and sustainable resource management.

2 Biobased and naturally, derived green solvents

Biobased and naturally derived solvents represent a rapidly growing category of green alternatives, offering low toxicity, high biodegradability, and derivation from renewable resources. These solvents are gaining widespread industrial application due to their reduced environmental impact and compatibility with green chemistry principles (Table 1). In recent years, the global chemical industry has increasingly prioritized renewable and non-toxic solvent systems to address concerns related to environmental sustainability, worker safety, and regulatory compliance. Biobased solvents have attracted particular attention because they originate from agricultural waste, forestry by-products, and fermentation processes, all of which align with the circular bioeconomy framework [152]. Their renewable origin reduces reliance on fossil fuels while simultaneously supporting carbon–neutral production cycles. These solvents also tend to exhibit high biodegradation rates, which significantly lowers environmental persistence compared with petroleum-derived solvents. As industries such as pharmaceuticals, agrochemicals, polymer processing, and food technology transition toward greener manufacturing practices, biobased solvents have become essential components of sustainable process design [78]. Furthermore, as consumers demand cleaner product labels and eco-friendly manufacturing practices, the adoption of naturally derived solvents

Table 1 Overview of selected biobased and naturally derived green solvents

Green solvent	Source	Applications	Features	Harmful effects of conventional solvents replaced	References
Limonene	Citrus fruit peels	Degreasers, polymer solvents, insecticides	Biodegradable, low toxicity, pleasant odor	Replaces toluene/xylene—neurotoxicity, air pollution	[120, 184]
Pine oil	Pine needles and twigs	Disinfectants, adhesives, solvents in paints	Antimicrobial, low aquatic toxicity, renewable	Replaces synthetic disinfectants—respiratory irritation	[72, 149]
Ethanol	Biomass fermentation	Pharmaceuticals, cosmetics, biofuels	Low toxicity, high volatility, miscible	Replaces methanol/acetone—toxicity, flammability	[32]
Butanol	Lignocellulosic biomass	Coatings, adhesives, phenolic extraction	Low volatility, renewable, safer handling	Replaces hexane—neurotoxicity, environmental persistence	[141, 206]
Choline chloride & glycerol (NADES)	Choline chloride & glycerol	Phytochemical extraction, enzyme media	Low volatility, biocompatible, customizable polarity	Replaces methanol—toxicity, explosive potential	[53, 81]
Sugar-based NADES	Sugars & organic acids	Extraction of antioxidants, cosmetics	Biodegradable, safe, customizable polarity	Replaces ethanol/methanol—skin/eye irritation, VOCs	[74, 105]
Methyl soyate	Soybean oil	Degreasers, paint removers, biodiesel	Low volatility, EPA Safer Choice, hydrophobic solvent	Replaces chlorinated solvents—carcinogenicity, ozone depletion	[29, 198]
Palm oil-derived methyl esters	Palm oil	Lubricants, surfactants, detergents	Biodegradable, RSPO-certified, oxidative stability	Replaces petroleum surfactants—poor biodegradability	[60, 207]

continues to rise across sectors where safety and environmental responsibility are mandatory considerations.

A key advantage of biobased solvents is the diversity of chemical structures obtainable from natural feedstocks. Plant-derived terpenes, fatty acid derivatives, sugar-based components, and fermentation-derived alcohols provide an extensive palette of physicochemical properties that can be tailored to specific industrial requirements. This variability offers opportunities to replace hazardous solvents in extraction, synthesis, purification, and cleaning operations. Several biobased solvents demonstrate solvent strengths, polarity ranges, and selectivity comparable to or exceeding those of traditional solvents such as dichloromethane, toluene, or hexane. Particularly promising are NADES, which offer tunable polarity and hydrogen-bonding characteristics, enabling highly selective extraction of phytochemicals [36]. These features underscore the growing potential of biobased solvents to enhance process performance while reducing overall environmental and health impacts.

2.1 Terpenes

2.1.1 Limonene

Limonene is a naturally occurring monocyclic terpene primarily extracted from the rinds of citrus fruits such as oranges and lemons. It has excellent solvency properties and is extensively used in industrial degreasers, paint strippers, and household cleaning products. Limonene is a biodegradable, low-toxicity solvent, and its pleasant citrus odor enhances user acceptability. In essential oil extraction, limonene has been shown to increase efficiency while reducing residual solvent contamination [120]. In the polymer

industry, it serves as an eco-friendly alternative to toluene and xylene for dissolving polystyrenes and rubbers. Additionally, limonene exhibits insecticidal and antibacterial properties, making it valuable in agriculture and personal care [184].

Beyond its traditional use as a cleaning agent and degreaser, limonene is increasingly recognized as an effective solvent for advanced material processing. Studies have demonstrated its utility in the dissolution and recovery of polystyrene waste, providing a sustainable method for recycling plastic materials that otherwise accumulate in landfills [85]. Limonene-based recycling processes not only reduce toxic solvent requirements but also contribute to the circularity of plastics, particularly in packaging industries. Its strong solvency for hydrophobic molecules makes it suitable for natural product extraction, particularly in food-safe applications where solvent residues pose health and regulatory concerns. Due to its natural origin, limonene residues are generally considered GRAS (Generally Recognized as Safe), making it advantageous in nutraceutical and food processing industries [117].

Research has also explored limonene's potential as a substitute for petroleum-derived solvents in synthetic chemistry applications, including cycloaddition reactions and polymerization processes. In these systems, limonene has demonstrated favorable reaction kinetics, reduced toxicity, and lower environmental impact compared with aromatic hydrocarbon solvents [23]. Furthermore, agriculture has benefited from the compound's inherent biological activities, as limonene-based formulations can function both as extraction solvents and as active components in pesticide delivery systems. With increasing emphasis on natural pesticide alternatives, limonene's dual solvent and bioactive roles provide unique opportunities for sustainable agrochemical development.

2.1.2 Pine oil

Pine oil, obtained via steam distillation of pine needles, twigs, and cones, is a complex mixture rich in terpene alcohols such as alpha-terpineol. It possesses strong antimicrobial, antifungal, and anti-inflammatory properties, making it suitable for disinfection in hospitals and clinical settings [149]. Pine oil is also used as a solvent in adhesives, paints, and cleaning formulations. It is considered safe for both users and the environment due to its rapid biodegradability and low aquatic toxicity [72]. Moreover, pine oil has potential for use in organic synthesis as a renewable reaction medium.

In addition to its disinfectant properties, pine oil has been evaluated for its solubilizing capabilities in various industrial processes, particularly as a replacement for petrochemical solvents in coatings, varnishes, and wood treatment applications. Owing to its high content of terpene alcohols and oxygenated monoterpenes, pine oil demonstrates excellent solvency for resins, natural gums, and hydrophobic organic compounds. This makes it a valuable green alternative for resin dissolution in furniture manufacturing and restoration industries [55]. Furthermore, pine oil's pleasant odor, low vapor pressure, and favorable safety profile provide occupational advantages over traditional solvents that may pose inhalation hazards or contribute to indoor air pollution.

Emerging research has also highlighted pine oil's potential in catalytic organic reactions, including esterification and oxidation processes. These applications leverage its polarity and hydrogen-bonding capability, providing reaction environments that reduce the need for hazardous solvents like dichloromethane and toluene. Pine oil's use in such reactions aligns well with green chemistry objectives, particularly in reducing waste

generation and improving reaction safety. Additionally, pine oil-based solvent blends have shown promise in environmental remediation, where their natural origin and low toxicity enable safe deployment for degreasing contaminated surfaces or solubilizing hydrophobic pollutants in soil and water systems [191]. As industries shift toward more sustainable manufacturing practices, pine oil's combination of antimicrobial efficacy, solvency performance, and environmental compatibility continues to position it as an increasingly valuable component of eco-friendly solvent systems.

2.2 Bioalcohols

2.2.1 Ethanol

Ethanol is a widely adopted green solvent derived from the fermentation of biomass such as corn, sugarcane, or lignocellulosic material. It is extensively used in pharmaceutical preparations, cosmetics, personal care products, and biofuels. Due to its relatively low toxicity, high volatility (allowing easy removal), and excellent miscibility with water and organic compounds, ethanol serves as a versatile solvent [32]. It is also biodegradable and renewable, with a favorable safety profile. According to the CHEM21 solvent selection guide, ethanol scores well in terms of health, safety, and environmental criteria.

Beyond its established role in pharmaceutical and cosmetic formulations, ethanol has gained significant importance as a food-grade solvent in nutraceutical industries, where strict toxicological limits prevent the use of petroleum-derived solvent residues. Its applicability spans tincture production, botanical extraction, traditional medicine standardization, and flavor and fragrance manufacturing. The tunable polarity achievable by altering ethanol–water ratios provides enhanced extraction selectivity for polyphenols, organic acids, flavonoids, and terpenoids [113]. This versatility has led to its widespread adoption in green extraction technologies such as ultrasound-assisted extraction, microwave-assisted extraction, and accelerated solvent extraction, where ethanol functions as an efficient medium capable of delivering high yields while minimizing process hazards.

Ethanol also plays a strategic role in analytical chemistry as an emerging substitute for acetonitrile and methanol in chromatographic separations, aligning with the principles of green analytical chemistry. With global bioethanol production increasing due to advancements in cellulosic fermentation and biorefinery integration, high-purity, cost-effective ethanol will become more widely available. Research has also shown that ethanol can be used as a co-solvent in catalytic reactions involving enzymes or metal complexes, improving solubility profiles while reducing toxicity associated with aprotic solvents [201]. Collectively, ethanol's broad acceptance across regulatory frameworks, combined with its low environmental footprint, underscores its increasing relevance as a sustainable and multifunctional solvent.

2.2.2 Butanol

Butanol, especially biobutanol derived from lignocellulosic waste, offers a higher energy content and lower vapor pressure than ethanol, making it more suitable for industrial use. It is used in coatings, adhesives, plasticizers, and biofuel formulations [141]. Butanol exhibits lower volatility and toxicity, reducing worker exposure and flammability risks. It also has higher hydrophobicity, which improves solubilization of nonpolar compounds. Recent research shows that bio-based butanol can serve as an effective extraction solvent for phenolics and lipids in food and pharmaceutical applications [206].

In addition to its established uses, butanol has gained attention within green chemistry research due to its intermediate polarity, which places it between alcohols like ethanol and nonpolar solvents such as hexane. This unique position on the polarity spectrum allows butanol to dissolve a wider range of chemical classes, including steroids, fatty acids, waxes, and certain polymeric materials, while still maintaining manageable safety profiles. The reduced vapor pressure of butanol not only lowers VOC emissions but also minimizes solvent loss during large-scale operations, improving overall process efficiency and reducing environmental burden [199].

Biobutanol's compatibility with both polar and nonpolar solutes has facilitated its application in green extraction processes, particularly in the isolation of lipophilic antioxidants, essential oils, and pharmaceutical intermediates. Its higher boiling point also enables selective fractionation and solvent recovery via energy-efficient distillation procedures. In catalytic processes, butanol can serve as a green reaction medium for esterification, transesterification, and biocatalytic transformations, often providing improved yields and selectivity relative to traditional solvents. Increasing interest in ABE (acetone–butanol–ethanol) fermentation technologies, combined with efforts to valorize agricultural and forestry residues, suggests that biobutanol production will continue to scale, enhancing its availability as a sustainable solvent for industrial applications [159].

2.3 Natural deep eutectic solvents (NADES)

2.3.1 Choline chloride and glycerol

NADESs composed of choline chloride and glycerol (typically 1:2 molar ratio) exhibit low volatility, high viscosity, and tunable polarity. These solvents are used for the extraction of polyphenols, alkaloids, and flavonoids from plants such as olive leaves and Scutellaria baicalensis [81]. They also serve as media for enzyme catalysis and pharmaceutical formulations due to their biocompatibility. Compared with methanol, choline chloride-glycerol NADESs have demonstrated higher extraction yields up to 94.7% for flavonoids [53] making them efficient and safer alternatives in botanical processing.

To further emphasize their growing importance, choline chloride–glycerol NADES have been extensively studied for their ability to stabilize sensitive biomolecules during extraction processes. Their unique hydrogen-bonding networks help prevent oxidative degradation of polyphenols and pigments, which normally degrade rapidly in alcoholic or petroleum-based solvents. Owing to the highly structured yet flexible nature of the NADES matrix, enzyme integrity is often preserved, enabling NADES-based systems to function as both solvents and biocatalytic microenvironments. This is particularly advantageous in pharmaceutical biotransformations where enzyme stability, solvent compatibility, and product purity are critical. Studies have shown improved enzyme activity and selectivity in NADES relative to aqueous or organic media, which opens opportunities for greener enzymatic synthesis pathways [71].

Recent work has also demonstrated that choline chloride–glycerol systems can enhance solubility and bioavailability of poorly water-soluble compounds. Their tunable polarity and low toxicity make them excellent carriers for antioxidant-rich extracts in nutraceutical and cosmetic formulations. Despite their high viscosity, NADES can be diluted with water to adjust handling characteristics without significantly compromising extraction performance. Their non-volatile nature also eliminates solvent emission concerns and minimizes risks associated with inhalation exposure in industrial settings.

Due to favorable thermal stability and biodegradability, these NADES systems represent a promising class of green solvents capable of addressing both environmental and functional limitations associated with conventional extraction solvents [192].

2.3.2 Sugar-based NADES

These NADESs typically involve sugars like glucose, fructose, or sucrose paired with acids (citric, lactic) or urea. They are used in cosmetics and food processing for the extraction of natural colorants, essential oils, and antioxidants. Sugar-based NADESs are nonflammable, inexpensive, and offer customizable polarity. Their biodegradability and low toxicity make them suitable for products in contact with human skin or ingestion [74, 105]. Recent studies indicate their superior solubilizing ability for phenolics compared to conventional ethanol or methanol-based solvents.

Beyond their use in extraction, sugar-based NADES have been evaluated for their ability to improve solubility and stability of heat-sensitive compounds such as anthocyanins, carotenoids, and sterols. These solvent systems mimic natural intracellular environments, particularly in plant tissues, making them excellent media for preserving molecular integrity throughout the extraction process. Their ability to form extensive hydrogen-bond networks enables strong interactions with phenolic and aromatic structures, often resulting in significantly improved extraction yields and reduced degradation [79]. As a result, sugar-based NADES have become attractive for producing high-quality extracts for functional foods, skincare products, and medicinal plant preparations.

Several studies have investigated their suitability as dispersing media, encapsulation agents, and carriers for active ingredients. Sugar-based NADES can enhance dermal permeation of bioactive compounds, which is beneficial in cosmetic formulations where deeper skin penetration is desirable. Additionally, their natural origin and edible constituents make them compatible with food-grade applications, avoiding regulatory complications associated with petroleum-derived solvent residues. In green synthesis, sugar-based NADES have been used as solvent media for nanoparticle fabrication, Maillard reaction studies, and enzymatic transformations, demonstrating versatility beyond extraction [122]. Their customizable polarity, low environmental burden, and broad compatibility with biological molecules underscore their emerging role as multifunctional green solvents in diverse industrial sectors.

2.4 Fatty acid methyl esters (FAMEs)

2.4.1 Methyl soyate

Methyl soyate is produced via transesterification of soybean oil and serves as a low-volatility, low-toxicity solvent in industrial degreasers, paint removers, and ink cleaners. It is recognized by the U.S. EPA's Safer Choice Program and exhibits high solvency for hydrophobic compounds [198]. Studies have demonstrated its efficacy in replacing chlorinated solvents in cleaning applications while maintaining performance. Methyl soyate also contributes to biodiesel formulations, offering a dual function as a fuel and solvent [29].

Methyl soyate has demonstrated exceptional potential as an environmentally benign alternative to petroleum-derived solvents due to its high boiling point, low vapor pressure, and strong solvency for nonpolar contaminants. Its ability to dissolve greases, oils, waxes, and polymeric residues makes it a preferred component in eco-friendly cleaning formulations for automotive, aerospace, and industrial equipment maintenance. Because

of its mild odor and biodegradability, methyl soyate improves workplace safety and reduces environmental impact when compared to volatile chlorinated solvents traditionally used for heavy-duty cleaning tasks. Additionally, its compatibility with surfactants and emulsifiers enables formulation of efficient water-based cleaning systems, which further reduce dependence on harmful solvent mixtures [137].

Advanced research has also evaluated methyl soyate as a medium for catalytic processes, such as esterification and transesterification, where it can act both as a solvent and as a reactant. In polymer chemistry, methyl soyate has been explored as a plasticizer and reaction medium for bio-based polymer synthesis, taking advantage of its relatively high polarity and strong hydrophobic interactions [29]. Due to its derivation from agricultural feedstocks, methyl soyate production contributes to rural economies and supports sustainable agro-industrial value chains. Its low toxicity toward aquatic and terrestrial organisms also makes it suitable for applications in environmental remediation, particularly in cleaning oil-contaminated surfaces and equipment [63]. These attributes position methyl soyate as one of the most commercially viable green solvents currently available.

2.4.2 Palm oil-derived methyl esters

Palm oil methyl esters are extensively used in surfactants, lubricants, and personal care products due to their emulsifying and solvent properties. These esters are fully biodegradable and can be produced from sustainably sourced palm oil, particularly that certified by the Roundtable on Sustainable Palm Oil (RSPO), which promotes responsible environmental and social practices in palm cultivation. They demonstrate effective grease removal and emulsification in detergent formulations [60, 207]. Their compatibility with other natural ingredients makes them ideal for eco-label products. Compared to soy-based FAMEs, palm-derived esters typically exhibit higher oxidative stability and are more suitable for tropical and industrial applications. However, sustainability concerns have been raised due to deforestation and habitat loss associated with palm plantations. To address this, industry practices increasingly favor certified sustainable palm oil to ensure reduced environmental impact. While palm oil-derived solvents can be considered green under certain conditions particularly when RSPO-certified and used in biodegradable, non-toxic applications their classification depends on both feedstock origin and production practices. As such, lifecycle assessments and sourcing transparency are essential when designating palm-based esters as "green solvents."

3 Green solvent replacement as a solution for environmental pollution

According to [199], there are two primary strategies for replacing conventional hazardous solvents: repurposing existing chemicals (route I) or developing entirely new, environmentally friendly solvents (route II). While both industry and academia actively pursue methods to accelerate the discovery of greener alternatives, the industrial sector generally favours route I because of its practicality and cost effectiveness. Conversely, academia often focuses on route II, which is typically supported by research grants or in collaboration with industry partners.

3.1 Real-world industrial applications of green solvents

Green solvents have progressed from laboratory-scale innovations to industrial-scale implementations across multiple sectors, demonstrating their practicality, economic feasibility, and regulatory alignment. Their adoption is often driven by both environmental regulations and the need to improve occupational safety and reduce waste generation.

One of the earliest and most widely adopted green solvent technologies is supercritical carbon dioxide (SC–CO_2) extraction, which is extensively used in the food and beverage industries. For instance, SC CO_2 is employed in the decaffeination of coffee and the extraction of hop resins and essential oils in brewing. Its advantages include high selectivity, the absence of toxic residues, and the preservation of thermolabile compounds due to mild extraction conditions [204]. Companies like Swiss Water Decaf and Hopsteiner have successfully scaled SC–CO_2 extraction systems to process tons of raw material annually, reducing reliance on conventional organic solvents like methylene chloride.

Natural deep eutectic solvents (NADES) and bioethanol have replaced hexane and petroleum ether in various botanical and herbal extraction facilities. Companies such as Indena S.p.A. in Italy and Gaia Herbs in the United States utilize NADES for the selective extraction of polyphenols, flavonoids, and alkaloids from plant matrices. NADES offer tunable polarity and are biodegradable, making them ideal for high-purity herbal APIs (active pharmaceutical ingredients) [89]. Bioethanol especially from renewable biomass has become a standard solvent for clean-label nutraceuticals and functional food ingredients, especially in ASEAN and EU-regulated markets.

Cyrene™ (dihydrolevoglucosenone), a bio-derived dipolar aprotic solvent developed by Circa Group in collaboration with the University of York, is increasingly replacing hazardous solvents such as dimethylformamide (DMF) in pharmaceutical research and polymer synthesis. Cyrene™ has been employed in reaction media, membrane fabrication, and graphene exfoliation, offering advantages in safety, renewability, and process performance [30]. It is currently used by multiple pharmaceutical companies in preclinical formulation work and green chemistry pilot studies.

Industrial biorefineries are beginning to adopt ionic liquids (ILs) and biobased solvents like γ-valerolactone (GVL) for lignocellulosic biomass fractionation and the catalysis of platform chemicals such as levulinic acid, furfural, and lactic acid [19]. GVL has gained traction due to its low toxicity and recyclability, and is now being evaluated for scale-up in biomass conversion plants in Europe and North America. These solvents provide an environmentally benign medium for catalytic reactions and are integral to circular bioeconomy frameworks.

In the coatings and surface cleaning sectors, methyl soyate and limonene are widely used as alternatives to xylene, toluene, and acetone. Automotive manufacturers such as Ford Motor Company have integrated soy-based solvents into degreasing operations and painting lines, resulting in reduced VOC emissions and improved worker safety [109]. Limonene, derived from citrus peel waste, is employed in resin cleaning and adhesive formulations, highlighting its valorization from agro-waste streams [24].

3.2 Alternative green solvents for hazardous compounds

3.2.1 Replacement of dichloromethanein extraction and reaction media

Chlorinated solvents such as dichloromethane (DCM) are widely used in extraction, chromatography, and organic synthesis due to their low boiling point, excellent solvency,

and easy removability. However, DCM is flagged as a Class 2 solvent under ICH guidelines because of its carcinogenic potential, volatility, and contribution to ozone layer depletion [158]. Consequently, greener substitutes are being explored across various applications.

Among potential alternatives, ethyl acetate is favored in pharmaceutical and botanical extractions for its moderate polarity, low toxicity, and biodegradability, although it may not always match DCM's performance in phase separation or solvent power [39]. Ethanol, especially in aqueous mixtures, has been utilized as a safer extraction solvent in nutraceuticals and food-grade preparations, but its higher polarity restricts its general substitutability for DCM in nonpolar systems [42]. Notably, solvents like dimethyl carbonate, 2-MeTHF, and Cyrene™ offer more structurally and functionally aligned replacements for DCM in synthetic chemistry workflows, especially for reactions requiring aprotic media or halogen-free conditions [158, 176].

It is critical to underscore that solvent substitution must consider both performance characteristics and health-environmental profiles. A solvent may be "greener" in classification but unsuitable for certain DCM-specific tasks, underscoring the need for case-specific selection criteria.

3.2.2 Natural product extraction and botanical applications

Natural product extraction is a critical step in the development of pharmaceuticals, nutraceuticals, cosmetics, and food additives. Traditionally, solvents such as hexane, chloroform, and dichloromethane have been employed for their high solvating power; however, their toxicity, volatility, and environmental persistence have driven efforts to identify greener alternatives.

Recent developments have demonstrated that bio-based solvents such as ethyl lactate, 2-MeTHF, d-limonene, and natural deep eutectic solvents (NADES) can achieve comparable or even superior extraction yields. For instance, Chemat et al. [42] highlighted the use of ethanol and ethyl acetate in replacing hexane in essential oil and polyphenol extraction, noting improved safety and comparable recovery efficiency. Similarly, Breil et al. [28] compared multiple green solvents, including *p*-cymene, 2-MeTHF, and CPME, for lipid extraction from *Yarrowia lipolytica*, concluding that these alternatives matched hexane in performance while offering environmental advantages.

Another notable development is the application of NADES, composed of natural constituents such as choline chloride and organic acids. These solvents demonstrate strong hydrogen bonding capacity, tunable polarity, and excellent biocompatibility. Studies show that NADES can extract flavonoids, alkaloids, and other secondary metabolites from plant matrices at yields comparable to methanol or ethanol, while being non-toxic and biodegradable [53, 151].

Overall, the use of green solvents in botanical extractions supports the development of safer, more sustainable supply chains for herbal medicines, cosmeceuticals, and food supplements, aligning with the principles of green chemistry.

NADES are gaining traction as efficient and eco-friendly alternatives in plant extraction. Composed of hydrogen bond donors and acceptors (e.g., choline chloride and lactic acid), NADES are biodegradable, tunable, and non-volatile. Dai et al. [53] demonstrated a choline chloride:lactic acid (1:2) NADES system yielding 94.7% flavonoid recovery from Scutellaria baicalensis, surpassing conventional methanol (78%). Cyrene™

(dihydrolevoglucosenone), a dipolar aprotic solvent derived from cellulose, has been successfully used in organic synthesis as a replacement for hazardous solvents like DMF and NMP. According to recent reports, Cyrene™ enables high-yield reactions with comparable or improved efficiency and selectivity, while offering reduced toxicity and enhanced biodegradability [46, 106]. γ-Valerolactone (GVL), derived from levulinic acid, is another renewable solvent with promising extraction capabilities. Studies show that GVL–water mixtures significantly improve phenolic compound solubility and extraction efficiency compared to ethanol, offering a safer and more sustainable alternative for natural product extraction [135]. The comparative performance of these novel green solvents is illustrated in Fig. 5.

3.2.3 Replacement of undesirable solvents in analytical chemistry

Recent advancements in green analytical chemistry have focused primarily on two key areas: the development of solvent-free extraction techniques and the substitution of hazardous solvents with more environmentally benign alternatives (Table 2) [185]. Despite these efforts, high-performance liquid chromatography (HPLC) has remained the benchmark method for analytical separations, where acetonitrile (CH_3CN) has traditionally been the solvent of choice [181, 201]. However, the cumulative solvent waste generated from multiple HPLC systems, even within a single institution or laboratory, is substantial. This has driven the search for greener alternatives to CH_3CN in routine analytical workflows [196]. Ethanol (EtOH) has emerged as a viable alternative to CH_3CN,

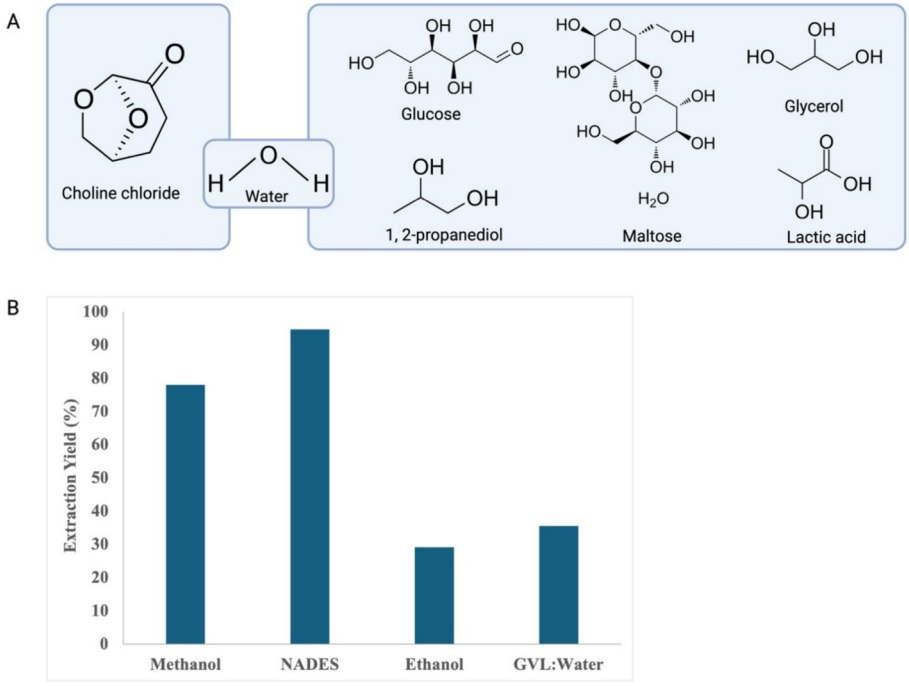


Fig. 5 Composition and comparative extraction efficiency of green solvents. **A** Representative molecular structures used in the preparation of natural deep eutectic solvents (NADES), consisting of choline chloride (hydrogen bond acceptor), water (as a ternary component), and various hydrogen bond donors including glucose, maltose, glycerol, 1,2-propanediol, and lactic acid. **B** Bar graph showing the extraction yield (%) of selected green solvents in specific plant extraction applications. NADES achieved the highest yield (94.7%), outperforming conventional methanol (78%). GVL:water (80:20) mixtures also demonstrated superior extraction (35.5%) compared to ethanol (29.1%). These findings highlight the enhanced efficiency of novel green solvents in natural product extraction [46, 106, 135]

Table 2 Possible alternatives to certain undesirable solvents

Undesirable solvent	Alternative solvent	Harmful effects of conventional solvent	References
Pentane or hexane(s)	Heptane	Neurotoxicity, suspected carcinogenicity, flammability	[8, 35, 100]
Ethers	2-MeTHF/Methyl *t*-butyl ether (MTBE)	Highly flammable, peroxide formation, CNS depression	[100]
Dichloromethane (extractions)	Toluene, MTBE, 2-MeTHF, Ethyl acetate	Suspected carcinogen, environmental toxicity, ozone depletion	[41, 100, 123, 205]
Dichloromethane (chromatography)	Heptane mixture/ethyl acetate	Suspected carcinogen, harmful vapors, potential reproductive toxicity	[41, 100]
Acetonitrile (HPLC mobile phase)	Ethanol–water mixtures	Toxicity, flammability, hazardous waste accumulation; classified as harmful and environmentally burdensome	[158]
Methanol (chromatography and extraction)	Bioethanol, ethyl lactate	High toxicity, blindness risk on exposure, high volatility and VOC contribution	[62]
Tetrahydrofuran (THF)	Cyclopentyl methyl ether (CPME), 2,2,5,5-tetramethyloxolane (TMO)	Forms explosive peroxides, strong CNS depressant, high flammability	[158, 180]
Toluene	Limonene, *p*-cymene, bio-based aromatic substitutes	Neurotoxicity, liver/kidney damage, persistent air pollutant	[142]
Petroleum ether	Ethanol–water mixtures, bioalcohols, terpenes	High flammability, VOC emissions, low biodegradability	[158, 203]
Dimethylformamide (DMF)	Cyrene™, propylene carbonate	Reproductive toxicity, hepatotoxicity, high environmental persistence	[66]
N-Methyl-2-pyrrolidone (NMP)	Gamma-valerolactone (GVL), ionic liquid alternatives	Reproductive toxicity (REACH Annex XVII), dermal penetration hazard	[40]
Hexane (oil extraction)	Ethanol, 2-MeTHF, ethyl acetate, isopropanol	Neurotoxicity, persistence, flammability, occupational exposure risk	[21]
Chloroform	Ethyl acetate, dimethyl carbonate	Probable human carcinogen, cardiotoxicity, high volatility, persistent atmospheric pollutant	[1]
Xylene	D-Limonene, pine oil fractions	CNS depression, respiratory irritation, environmental dispersion	[98]
Ethyl ether	2-MeTHF, CPME	Auto-oxidation forming explosive peroxides, high volatility	[15]

especially after optimizing chromatographic conditions such as flow rates [201]. Ethanol is characterized by lower toxicity and reduced vapour pressure compared to commonly used aprotic solvents such as acetonitrile and dichloromethane, making it a safer and more environmentally benign alternative in many extraction and chromatographic applications [75, 158]. The increasing global production of bioethanol is expected to further increase both the affordability and purity of EtOH in the foreseeable future [100].

From a chromatographic standpoint, EtOH offers properties comparable to those of MeOH and CH_3CN. Miyabe [133] reported similar adsorption mechanisms in reversed-phase HPLC (RP-HPLC) systems when EtOH/water, ACN/water, or MeOH/water mixtures were employed as mobile phases. Similarly, in a study by Ribeiro et al. [163] demonstrated that EtOH can successfully replace MeOH or CH_3CN as a greener solvent without compromising chromatographic efficiency. According to Snyder's solvent classification system, EtOH and MeOH fall within the same selectivity group, suggesting analogous solvent behaviors [114]. Moreover, EtOH provides greater eluotropic strength than MeOH does, allowing for lower solvent percentages in mobile phases to achieve equivalent retention times [175].

3.2.4 Replacement of green's solution in oil extraction

In a study by Li et al. [118] evaluated several solvents, such as hexane, ethanol, isopropanol, and terpenes, for their efficiency in extracting rapeseed oil. Among the alternatives, *p*-cymene (a terpene) yielded extracts with relatively high levels of polar lipids but relatively low concentrations of tocopherols and tocotrienols, suggesting that hexane could be replaced by *p*-cymene due to its favourable selectivity and lipid recovery. Similarly, Sicaire et al. [177] investigated 2-methyltetrahydrofuran (2-MeTHF) as a green substitute for hexane in the extraction of edible oil from rosemary seeds. When extraction was conducted at a 1:5 solvent-to-seed ratio at 55 °C for 8 h, the authors reported that 2-MeTHF-extracted oils contained higher concentrations of beneficial bioactive compounds, such as tocopherols, sterols, and tocotrienols, than hexane extracts did.

In a study by Breil et al. [28] expanded the investigation of green solvents for lipid extraction from Yarrowia lipolytica biomass using a 30-min maceration protocol. The study evaluated multiple candidates, including *p*-cymene, 2-methyltetrahydrofuran (2-MeTHF), and cyclopentyl methyl ether (CPME), alongside conventional green solvents such as ethanol, isopropanol, α-pinene, ethyl lactate, D-limonene, ethyl acetate, dimethyl carbonate, and isopropyl alcohol (IPA). Among these, only 2-MeTHF, ethyl acetate, and CPME demonstrated effective oil solubilization capabilities, comparable to hexane, while maintaining similar energy demands and carbon footprints. IPA was found to be less effective due to its limited oil solubility.

While 2-MeTHF has been extensively discussed in literature for its lipid recovery efficiency and low toxicity, CPME has emerged as a highly promising green solvent with additional industrial advantages. Its low peroxide formation, hydrophobicity, and chemical inertness make it particularly suited for lipid extractions involving unsaturated or oxidation-sensitive fatty acids. Notably, CPME's partial immiscibility with water facilitates downstream phase separation and solvent recovery, reducing the overall process burden. The boiling point (~106 °C) of CPME allows for energy-efficient evaporation, while its low miscibility with polar phases helps prevent emulsion formation during extraction an advantage over many alcohol-based solvents.

In recent studies, CPME has also been employed successfully in other green extraction platforms, including phytochemical isolation and essential oil recovery, due to its chemical stability and low toxicity profile [209]. In the context of oil extraction, CPME not only matched hexane in performance but did so with a significantly improved environmental and safety profile, aligning with the principles of green chemistry and industry goals for sustainable processing.

In a related investigation, Anisuzzaman and Yih [14] evaluated ethanol and isopropanol for the extraction of rice bran oil as potential hexane substitutes. While alcohols are typically linked with lower extraction efficiencies, isopropanol achieved higher tocopherol yields a key antioxidant in rice bran oil than hexane and other tested solvents, indicating that short-chain alcohols may outperform hexane in certain applications. Furthermore, Gharby et al. [77] studied oil extraction from *Opuntia ficus-indica* (cactus seeds) using 2-MeTHF, reporting a higher extraction yield (9.00 g/100 g) than hexane (8.86 ± 0.25 g/100 g). Oils extracted with 2-MeTHF also exhibited enhanced sterol content, lower acidity, and reduced peroxide values, underlining its suitability as a green and efficient alternative in oil extraction.

3.2.5 Ethanol and water as safer alternatives to hexane in natural product extraction

Hexane, a common petrochemical solvent used in oil and plant extractions, is associated with significant health and environmental concerns, including neurotoxicity, high volatility, and low biodegradability. According to the CHEM21 solvent selection guide, hexane is categorized as "hazardous and not recommended" due to its flammability, persistence, and toxicity (Table 3). In contrast, ethanol and water are both rated "recommended", based on their low toxicity, renewability, and favorable environmental profiles [158].

In a study by Akintobi et al. [7] investigated the antimicrobial activity of polyphenolic extracts (40 mg/mL) derived from *Parkia biglobosa* root bark using various solvents, including ethanol, water, and petroleum ether. This study revealed that only the hexane extract selectively inhibited *Klebsiella pneumoniae*, whereas the ethanol extract demonstrated broad-spectrum antibacterial activity against all the tested pathogens. Phytochemical analysis revealed the presence of terpenoids, alkaloids, flavonoids, tannins, saponins, steroids, phlobatannins, and anthraquinones in the root bark. However, anthraquinones could not be extracted by any of the solvents employed in the study.

In a study by Sandhya et al. [168] examined the extraction of polyphenols from pomegranate peels via aqueous and ethanolic solvents at two solvent-to-peel ratios (15:1 and 60:1). These findings showed that ethanol consistently outperformed water in terms of polyphenol yield across both ratios. Additionally, they reported that increasing the solvent-to-peel ratio significantly improved the extraction efficiency of polyphenols. In this study, ethanol–water mixtures yielded higher flavonoid and tannin recovery than water alone, highlighting how even within the green solvent domain, formulation choices can significantly affect outcomes.

In another study, Awan et al. [18] compared the extraction efficiencies of hexane, ethanol, and ethyl acetate in obtaining bioactive compounds from garlic (*Allium sativum*) at different extraction times (35, 50, and 65 min). Their results indicated that methanol (used interchangeably with ethanol in some studies) achieved the highest concentrations of phenolics (approximately 60 mg gallic acid equivalent per 100 g) and flavonoids (60 mg per 100 g) at an optimal extraction time of 50 min.

In a study by Torgbo et al. [187] explored the extraction of polyphenols from rambutan peel via ohmic heating combined with the addition of deionized water and ethanol. The study revealed that ethanol yielded a significantly higher polyphenol concentration than did water. The key bioactive components identified in the extract included corilagin, ellagic acid, and gallic acid. Furthermore, the ethanolic extract was successfully applied to extend the shelf-life of bread, resulting in antifungal activity that prolonged freshness from three to 7 days. These findings collectively highlight the potential of ethanol and other green solvents to serve as effective alternatives to petroleum-derived solvents in the processing of herbs and spices, offering improved extraction efficiency and added functionality in food preservation.

3.3 Newly discovered green solvents

Recent advancements in green chemistry have led to the discovery of novel solvents such as Cyrene™ and 2,2,5,5-tetramethyloxolane (TMO), which were developed through collaborative research efforts spearheaded by the Biobased Industries Consortium. Both

Table 3 Comparative evaluation of hexane, ethanol, and water using CHEM21 criteria

Solvent	Health (H) score	Health notes	Environment (E) score	Environment notes	Safety (S) score	Safety notes	Overall assessment	CHEM21 category
Hexane	5	Neurotoxic, possible carcinogen	4	Not readily biodegradable	5	Highly flammable, volatile	Poor	Not recommended
Ethanol	2	Low acute toxicity	2	Biodegradable, renewable source	2	Moderately flammable	Good	Recommended
Water	1	Non-toxic	1	Ideal—no environmental harm	1	Non-flammable	Excellent	Recommended

The table presents health, environmental, and safety scores alongside qualitative assessments. Hexane scores poorly, while ethanol and water are safer, greener alternatives recommended by the CHEM21 framework [158]

solvents are currently in the process of being upscaled for commercial production, offering new environmentally friendly alternatives to conventional solvents [210].

Cyrene™, an aprotic dipolar solvent derived from biomass waste, is based on dihydrolevoglucosenone and presents a promising green alternative to traditional dipolar aprotic solvents. It is characterized by low toxicity, biodegradability, and favourable solvation properties, making it a competitive substitute for petrochemical-based solvents [38]. The Green Chemistry Centre of Excellence at the University of York, in collaboration with the Circa Group, has actively promoted Cyrene™ as a sustainable alternative to widely used solvents such as *N*,*N*-dimethylacetamide (DMA), *N*,*N*-dimethylformamide (DMF) [176]. Research has subsequently demonstrated the versatility of Cyrene™ across various industries, with promising applications in multiple chemical processes [37]. Notably, Cyrene™ and its aqueous mixtures have shown efficacy in facilitating a range of organic reactions and have been applied in the development of advanced materials such as polymeric membranes, graphene, and metal–organic frameworks [57].

In a study by Trowse et al. [188] evaluated the performance of TMO, a biobased solvent with physicochemical properties similar to those of toluene, and found it to be effective in Buchwald–Hartwig amination reactions, facilitating the coupling of both primary and secondary amines with aryl bromides. These findings align with those of [153], who identified TMO as a safer and greener substitute for solvents such as tetrahydrofuran and toluene (Table 4) in the biocatalyzed synthesis of polyesters. Overall, the emergence of Cyrene™ and TMO underscores the growing portfolio of biobased solvents designed to reduce the environmental and health hazards associated with conventional solvents while maintaining or enhancing process efficiency.

4 Advances, innovations and industry developments in green solvents

Significant advances in green chemistry have led to the development of environmentally friendly innovations, particularly within research and development (R&D) sectors. Between 1983 and 2001, approximately 3200 patents were issued in the United States, primarily targeting the chemical industry. A considerable portion of these patents focus on green chemistry technologies, notably those related to solvents and polymers. Both governmental bodies and major industrial stakeholders have increasingly prioritized the adoption of green chemistry practices to meet regulatory and environmental demands. Since the 1980s, the U.S. government has repeatedly revised regulations to encourage green chemistry implementation, particularly in response to growing environmental concerns and public health risks associated with hazardous chemicals. Today, with the steady rise in green chemistry-related patents, regulatory frameworks have evolved to ensure that green solvents are not exploited as mere regulatory loopholes but are genuinely integrated into sustainable industrial practices.

4.1 Green solvent applications in agrochemicals

Green solvents have demonstrated considerable potential in agrochemical applications, offering improved environmental and safety profiles. Typically comprising natural oils (70–99%), cosolvents (0.1–20%), and antifreeze agents (up to 10%), these formulations reduce toxicological burdens. For instance, Musarurwa and Tavengwa [136] described a green pesticide solvent system utilizing gemini pyrrolidone, demonstrating superior solubility and stability compared to benzene. Biobased herbicidal formulations utilize

Table 4 Possible alternative to newly discovered green solvents

Green solvents	Possible Hazardous Substances that can be substituted	References
Cyrene™	Existing dipolar aprotic solvents, such as *N*,*N*-dimethylacetamide (DMA), *N*,*N*-dimethylformamide (DMF), and *N*-methyl pyrrolidinone (NMP)	[176]
TMO	Toluene Tetrahydrofuran	[33, 34, 153]

renewable green solvents such as bioethanol, limonene, and fatty acid methyl esters. These systems provide effective weed control while minimizing ecological harm [84]. Similarly, biostimulant delivery systems use biodegradable polymer matrices or lipid-based carriers with green solvents to enhance plant nutrient uptake, reduce leaching, and improve soil and water quality [6, 165]. Environmentally friendly pesticide encapsulation methods further integrate green solvents such as glycerol or terpenes to enhance the controlled release and stability of active ingredients. Encapsulation reduces runoff, improves field efficacy, and minimizes application frequency, thereby lowering ecological impact [161].

4.2 Green solvents in natural product extraction

The use of green solvents in the extraction of natural products has gained considerable momentum in pharmaceutical, food, nutraceutical, and cosmetic industries. Supercritical CO_2 is a non-toxic, non-flammable solvent widely used for high-purity extraction under mild conditions [42]. It provides high selectivity without leaving toxic residues. Ionic liquids (ILs), characterized by low volatility and tunable properties, enable efficient extraction of both polar and nonpolar compounds from diverse natural matrices while reducing solvent waste and allowing for recycling [145]. Plant-derived green solvents, such as those from turpentine, offer renewable, low-toxicity alternatives rich in pinene, limonene, and camphene. These solvents are prepared through destructive distillation and fractionation of rosin, and have proven safe and effective for pesticide and fragrance extraction [5, 125].

4.3 Green solvents in polymer and waste management

Green solvents are also revolutionizing polymer recycling and waste oil treatment. Supercritical fluid-assisted depolymerization of polyethylene terephthalate (PET) enables recovery of monomers under mild conditions, supporting high-quality recycled PET (rPET) production [76]. Solvent-based recycling processes selectively dissolve and purify polymers from mixed plastic waste streams, facilitating closed-loop recycling and minimizing landfill accumulation [93]. Ionic liquids have also been explored for polystyrene (PS) recycling, dissolving the polymer into oligomeric fractions with minimal energy input and hazardous waste generation [44]. Waste lubricating oils (WLO) can be treated using green extractants (e.g., C3–C5 alcohols and ethanolamine), achieving >92.5% oil yield and >85.5% solvent recovery. Comparative studies found 1-butanol more effective than MEK in sludge removal across various WLO types [111, 155].

4.4 Innovations in industrial and specialty applications

In the extraction of montmorillonite wax from brown coal, green solvent mixtures of cyclohexane and alcohol improve efficiency while reducing toxicity. The process involves crushing lignite, extraction under heat and pressure, flash evaporation, and final

refinement [208]. Pigment and dye manufacturing have also benefited [157]. A method for synthesizing phthalocyanine green pigment enhances stability and reduces flocculation in inks through pH adjustment and multi-step filtration [167]. Similarly, a low-pollution route to solvent orange 60 uses NMP but claims eco-friendliness due to process optimization [121]. Green detergents using halloysite nanotubes (HNTs) enable effective emulsion stabilization and stain removal, outperforming traditional surfactant-based formulations while maintaining environmental compatibility [202]. Together, these advances demonstrate the expanding role of green solvents across agrochemical, natural product, waste treatment, and specialty industrial applications, highlighting the versatility and impact of sustainable solvent systems.

5 Challenges regarding green solvents

Although green solvents offer significant environmental and health advantages, several challenges limit their widespread adoption in industrial and research settings. One of the major bottlenecks is the limited availability of comprehensive toxicological and ecotoxicological data for many recently introduced solvents, particularly NADES, ionic liquids, and next-generation bio-derived solvents such as Cyrene™ and TMO [172]. Without long-term environmental fate profiles or robust biodegradation studies, regulatory approval becomes difficult, creating uncertainty for manufacturers seeking to replace established solvents.

Cost and scalability present additional challenges. While solvents like ethanol and methyl soyate are economically viable, many newly developed green solvents require specialized synthesis pathways or rely on limited raw materials. These factors elevate production costs and restrict large-scale supply. Furthermore, differences in physical properties such as viscosity, boiling point, and polarity often require redesigning existing processes or equipment. High-viscosity NADES, for example, can complicate pumping, mixing, and filtration operations [9].

Another challenge is the variability in sustainability of the feedstocks used to produce green solvents. For instance, the environmental impact of palm oil-derived esters largely depends on land-use practices and certification. The "greenness" of these solvents is therefore not intrinsic, but dependent on sustainable agricultural practices, traceability, and supply chain certifications. Additionally, solvent recovery can be more difficult for some green solvents, particularly those with low volatility. Yield losses, energy intensity, and complex separation steps can offset environmental benefits [162].

Finally, industries may be reluctant to transition without standardized assessment metrics. While tools such as CHEM21 assist in solvent selection, differences in regional regulatory frameworks and the absence of universally accepted greenness scoring systems complicate decision-making. Collectively, these challenges highlight the need for more research, regulatory clarity, and infrastructure development to fully realize the benefits of green solvents [104].

6 Future prospects and successful case studies

Future developments in green solvents are closely aligned with advances in biorefinery technologies, computational solvent design, and circular economy principles. As biomass conversion becomes more efficient, the availability of renewable solvent feedstocks such as levulinic acid, glycerol, and lignocellulosic sugars is expected to significantly

increase [192]. This will lower production costs and expand the diversity of viable green solvents. Machine learning and molecular modeling are also emerging as powerful tools for solvent design, enabling prediction of solvent performance, toxicity, and compatibility based on molecular features. This computational approach is likely to accelerate the development of tailor-made solvents for specific industrial processes [107].

Several real-world success stories demonstrate the growing feasibility of green solvent adoption. Supercritical CO_2 extraction is now fully established in the decaffeination of coffee and the extraction of hops and essential oils, offering residue-free, high-quality extracts. Cyrene™, a cellulose-derived dipolar aprotic solvent, has been successfully implemented in polymer research, graphene exfoliation, and pharmaceutical synthesis as a replacement for DMF and NMP [87]. In the automotive sector, methyl soyate-based cleaners have replaced chlorinated solvents in degreasing operations, significantly reducing hazardous emissions. NADES have been adopted in commercial botanical extraction, providing higher yields and safer extracts for nutraceuticals and cosmetics.

Emerging evidence also supports the use of gamma-valerolactone in biomass fractionation, enabling efficient conversion of lignocellulose into platform chemicals while allowing solvent recycling. Furthermore, terpene-based solvents such as limonene and *p*-cymene have gained traction in the coatings and adhesive industries where low toxicity and pleasant odor are valued [173]. These successful adoptions illustrate that green solvents are not limited to academic research but are increasingly becoming integral components of industrial process optimization, environmental compliance, and product safety enhancement.

7 Conclusion

Green chemistry, particularly through the adoption of green solvents, presents a transformative pathway to reduce waste generation, enhance public and occupational health, and address environmental degradation. This review highlights the potential of biobased and naturally derived solvents such as terpenes, bioalcohols, natural deep eutectic solvents, and fatty acid methyl esters not only as safer alternatives to conventional organic solvents but also as contributors to sustainable industrial development. Their alignment with multiple Sustainable Development Goals reinforces their significance in driving eco-conscious innovation across pharmaceutical, agrochemical, polymer, and food industries. While the growing body of research and patent data reflects increasing global interest and investment, the field remains limited by several challenges. These include insufficient data on long-term biodegradation pathways, limited scalability and cost-effectiveness for certain novel solvents, and the need for standardized evaluation metrics such as unified greenness scores. Additionally, lifecycle assessments for emerging solvents particularly natural deep eutectic solvents and palm-based esters are still lacking or incomplete, which complicates efforts to classify these solvents uniformly as "green." Future research should focus on developing comprehensive toxicity and environmental fate profiles for new green solvents, conducting comparative performance trials against conventional solvents in real-world industrial scenarios, and expanding the availability of certified sustainable feedstocks (e.g., RSPO-certified palm oil). Advancing solvent recycling and recovery technologies will be crucial to enhance the economic feasibility of green solvent use. Furthermore, regulatory clarity and policy incentives are essential to accelerate the adoption of green solvents across industrial sectors. Ultimately,

transitioning to green solvents is not a standalone solution but a critical component of a broader commitment to green chemistry. When coupled with sustainable process design and cross-sector collaboration, this transition offers a tangible route toward a safer, greener, and more resilient future.

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Author contributions
P. V. Kumar had the idea for the article. MAA Maki, MTS Yee, and Tan Kin Fai performed the literature search and data analysis. MAA Maki, MP Venkatesh, DK Mishra, MK and OW Abduljaleen drafted the work. P. V. Kumar reviewed and edited the manuscript. All authors critically revised the manuscript.

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Declarations

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